MABEL Actuator Friction & Inertia Calibration

Identifying the Coulomb / viscous / Stribeck friction and reflected inertia of the Damiao FOC joints, for inverse dynamics, gravity compensation, and sim-to-real

MABEL calibration

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The companion package `mabel_sysid` identifies the *rigid-body* parameters of the arm (link mass and centre of mass) and explicitly *zeroes* joint friction and damping so the regression stays linear. On the real robot those zeroed terms are exactly what is left over: the brushless-DC pancake motor and its planetary gearbox add dry (Coulomb) friction, viscous damping, a velocity-dependent Stribeck dip, and reflected rotor inertia. Ignoring them is the dominant reason a feed-forward torque that is "correct" on paper still fails to move the joint, or a gravity-compensation hold drifts. This note derives the friction/inertia model, the experiments that identify it, and how the result plugs straight back into the MuJoCo model and the controller. The whole pipeline is first *proven in simulation* (§5), exactly as the rigid-body identification is.

Quantity	Symbol	MuJoCo joint attribute
Coulomb (dry) friction	F_c	`frictionloss`
viscous damping	F_v	`damping`
reflected rotor inertia	J	`armature`
break-away / Stribeck	F_s, v_s	(feed-forward only)

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1 Why this is needed

The Euler–Lagrange model used for inverse dynamics and gravity compensation is

$$M(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} + g(q) = \tau_{\mathrm{rb}}, \tag{1}$$

where τ_{rb} is the *rigid-body* joint torque. The torque the motor must actually command is

$$\tau_{\mathrm{cmd}} = \underbrace{M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q)}_{\tau_{\mathrm{rb}}\ \text{(from } \texttt{mabel_sysid}\text{)}} + \underbrace{\tau_{\mathrm{f}}(\dot{q})}_{\text{friction}} + \underbrace{J\,\ddot{q}}_{\substack{\text{reflected}\\ \text{inertia}}}. \tag{2}$$

Two practical consequences:

- **Gravity compensation.** A static hold commands $g(q)$. If $|g(q)| < F_{\mathrm{s}}$ the joint sticks; if you measure $g(q)$ from the holding torque, that measurement is biased by static friction. Friction must be characterised to both *trust* the gravity-torque measurements and *compensate* them.
- **Sim-to-real.** MuJoCo applies τ_{f} and J through the joint attributes `frictionloss`, `damping`, `armature`. With these left at default the simulator is frictionless and under-damped; identified values make the simulated joint behave like the real one.

2 The Damiao actuator and how we read torque

MABEL's arm uses Damiao quasi-direct-drive servos (`DM-J10422P` torso, `DM-J8009P` shoulders, `DM-J4340` elbows, `DM-J3507` wrists/neck), each a field-oriented-control (FOC) brushless-DC pancake motor with a low-ratio planetary gearbox. They are commanded over CAN at 1 Mbit in *MIT mode*: the host sends a frame packing $(p_{\mathrm{des}}, v_{\mathrm{des}}, k_p, k_d, \tau_{\mathrm{ff}})$ and the motor runs the internal impedance law

$$\tau = k_p\,(p_{\mathrm{des}} - p) + k_d\,(v_{\mathrm{des}} - v) + \tau_{\mathrm{ff}}. \tag{3}$$

The feedback frame returns position, velocity, and a *torque estimate* (output side, in N m) derived from the q-axis current and the motor torque constant. This estimate is the electromagnetic torque the motor applied; at constant velocity it equals the load it is fighting, which is the basis of the friction measurement below. (Driver: `src/mabel_actid/damiao.py`; field packing mirrors the firmware in `firmware/openarm`.)

Choosing k_p, k_d selects the experiment:

mode	gains	use
position ramp	$k_p > 0,\ k_d > 0$	hold a constant velocity accurately (friction)
velocity loop	$k_p = 0,\ k_d > 0$	simple velocity hold (drifts at low speed)
pure torque	$k_p = k_d = 0$	open-loop torque replay

3 The model

3.1 Friction

We use the classical Coulomb + viscous + Stribeck curve, with *direction asymmetry* (a planetary gearbox is rarely symmetric: backlash, helical thrust, and seal drag differ between drive directions):

$$\tau_{\mathrm{f}}(\dot{q}) = \left[F_{\mathrm{c}}^{\pm} + (F_{\mathrm{s}}^{\pm} - F_{\mathrm{c}}^{\pm})\, e^{-|\dot{q}/v_{\mathrm{s}}|^{\delta}}\right] \mathrm{sgn}(\dot{q}) \;+\; F_{\mathrm{v}}\, \dot{q}, \tag{4}$$

where the $\pm$ superscript selects the forward/reverse value by $\mathrm{sgn}(\dot{q})$. The terms are: $F_{\mathrm{c}}^{\pm}$ Coulomb (dry) friction; F_{v} viscous damping; $F_{\mathrm{s}}^{\pm}$ the break-away (static) torque at zero speed; v_{s} the Stribeck characteristic velocity over which static decays to Coulomb; and δ the shape exponent ($\delta = 2$, Gaussian). Special cases recover simpler models: $F_{\mathrm{s}} = F_{\mathrm{c}}$ and symmetry give plain Coulomb+viscous (the model MuJoCo can represent exactly); $F_{\mathrm{v}} = 0$ gives dry friction only. In code sgn is smoothed by $\tanh(\dot{q}/\varepsilon)$ so the model is differentiable for the optimiser.

3.2 Reflected inertia and the full torque

Adding the rigid-body inertia of the joint gives, for a single actuator,

$$\tau_{\mathrm{cmd}} = J\,\ddot{q} + \tau_{\mathrm{f}}(\dot{q}) + \tau_{\mathrm{g}}(q), \tag{5}$$

with $J = N^2 J_{\mathrm{rotor}} + J_{\mathrm{gearbox}} + J_{\mathrm{load}}$ the inertia seen at the *output* (N is the gear ratio), and τ_{g} any external (gravity) load. On a bare bench shaft $\tau_{\mathrm{g}} = 0$ and J is the actuator's own reflected inertia, which is precisely MuJoCo's `armature`.

4 Identification

The model is fit in two stages, one per excitation. Code: `src/mabel_actid/identify.py`.

4.1 Stage 1 — friction curve at constant velocity

At constant velocity $\ddot{q} = 0$, so (5) collapses to

$$\tau_{\mathrm{cmd}} = \tau_{\mathrm{f}}(\dot{q}) + \tau_{\mathrm{g}}(q). \tag{6}$$

On a bench with $\tau_{\mathrm{g}} = 0$ the measured torque *is* the friction. We drive a **velocity staircase**: a sequence of constant speeds $\{\dot{q}_i\}$, both directions, dwelling at each long enough to reach steady state, and average the reported torque over the settled part of each step. This traces out $\tau_{\mathrm{f}}(\dot{q})$ point by point — the most robust friction measurement there is.

Three design choices make this work in practice:

1. **Log-spaced speeds.** The Stribeck dip and break-away live at low speed, so the levels are geometrically spaced from $\sim$0.03 rad/s up to a safe fraction of $v_{\max}$, densely sampling the low end.
2. **Hold velocity with a position ramp** ($k_p > 0$), advancing p_{des} at the target speed, rather than a pure velocity loop. The Damiao k_d is capped ($k_d \leq 5$), so a velocity loop has a steady-state error $\approx \tau_{\mathrm{f}}/k_d$ that is ruinous near zero speed; a position loop holds slow speeds accurately. Interleaving $+\dot{q}_i, -\dot{q}_i$ keeps the ramp position bounded around the start.

3. **Reject stick-slip.** In the negative-slope (Stribeck) region the constant-velocity motion is unstable and breaks into a stick-slip limit cycle, so a few of the lowest-speed points have no well-defined steady friction. Two guards remove them: a convergence gate keeps only samples where the measured velocity has actually reached the command, and the curve fit uses a robust (`soft_l1`) loss that down-weights the remaining outliers. The robust loss is scaled by the measurement noise estimated from the clean high-speed residual.

Fitting. For the symmetric/asymmetric Coulomb+viscous models the fit is linear least squares on the regressors $[\operatorname{sgn}\dot{q},\ \dot{q}]$ (or $[\mathbf{1}_{\dot{q}>0},\ \mathbf{1}_{\dot{q}<0},\ \dot{q}]$). For the full Stribeck model it is a nonlinear least squares, *staged*: above the dip the curve is pure Coulomb+viscous, so a linear fit to the high-speed points robustly pins F_c, F_v; F_s, v_s are then seeded from the low-speed torque excess and refined. Staging plus bounding v_s inside the sampled speed range avoids the degenerate minimum (a blown-up v_s that flattens the exponential) a naive single-stage fit falls into.

4.2 Removing a gravity load: the ± trick

If the bench has a position-dependent load (e.g. a pendulum about a horizontal axis), run each speed at $+\dot{q}$ and $-\dot{q}$ at the *same position band*. Since friction is odd in $\dot{q}$ and the load is the same,

$$\tfrac{1}{2}\big[\tau(+\dot{q}) - \tau(-\dot{q})\big] = \tau_f(+\dot{q}), \qquad \tfrac{1}{2}\big[\tau(+\dot{q}) + \tau(-\dot{q})\big] = \tau_g(q). \tag{7}$$

The difference isolates friction; the sum estimates the load (`identify.gravity_split`).

4.3 Stage 2 — reflected inertia from acceleration

Constant-velocity data cannot see J (it multiplies $\ddot{q} = 0$). We add a **frequency chirp** in position, $q(t) = A\sin\big(2\pi(f_0 t + \tfrac{1}{2}kt^2)\big)$, sweeping $f_0 \to f_1$ to excite a broad acceleration spectrum so J is well conditioned. With the friction curve now known, the only unknown left in (5) is J:

$$\tau_{cmd} - \tau_f(\dot{q}) = J\,\ddot{q} \quad\Longrightarrow\quad J = \frac{\ddot{q}^\top\big(\tau_{cmd} - \tau_f(\dot{q})\big)}{\ddot{q}^\top\ddot{q}}, \tag{8}$$

a one-parameter linear least squares (optionally augmented with a $\sin q$ column to also fit a pendulum load). Acceleration is obtained from the measured velocity by a Savitzky–Golay filter (smooth-then-differentiate); raw finite differences of quantised velocity are unusable.

5 Simulation validation

Following the project's "prove it in sim first" rule, `act_00_simulate_check.py` builds a ground-truth actuator (a velocity-loop FOC plant with known friction and inertia, plus torque-estimate noise and velocity quantisation), generates a staircase and a chirp, runs the *identical* identification, and compares. Representative result for a `DM-J4340`-class joint (true $F_c^+ = 0.45$, $F_c^- = 0.52$, $F_v = 0.08$, $v_s = 0.15$, $J = 0.012$, torque noise $0.02\,\mathrm{N\,m}$):

parameter	true	identified	error
F_c^+ N m	0.4500	0.4501	0.0001
F_c^- N m	0.5200	0.5186	0.0014
F_v N m s	0.0800	0.0806	0.0006
v_s rad/s	0.1500	0.1650	0.0150
J kg m^2	0.0120	0.0120	0.0000

The three sim-to-real terms (F_c, F_v, J) are recovered to the noise floor: friction inlier RMS 0.0024 N m, inertia RMS 0.0435 N m, $R^2 = 0.93$ (depressed only by the deliberately rejected stick-slip outliers). Plots are written to `experiments/`: `act_friction_curve.png` (measured points, fitted and true curves) and `act_torque_fit.png` (chirp torque, measured vs model).

Honest limitation. The forward break-away F_s^+ is the one weak parameter: the very low-speed points that would reveal it are exactly those lost to stick-slip, so static friction is only partially identifiable from constant-velocity data. This does not affect F_c, F_v, J (the controller and MuJoCo terms), which are pinned by the well-behaved mid/high-speed data.

6 Procedure on hardware

1. **Mount** the motor on the bench with the *output shaft free* — no link, or a known, balanced load — so $\tau_g = 0$ and the measured torque is pure friction. Bring up CAN: `sudo ip link set can0 up type can bitrate 1000000`.
2. **Collect** (`act_01_collect.py --motor DM-J4340 --id 1`): runs the velocity staircase then the chirp, enabling on start and always disabling on exit. Start with a low `--vel-frac` and increase once you trust the rig. Writes `act_<id>_staircase.csv` and `act_<id>_chirp.csv`.
3. **Identify** (`act_02_identify.py --staircase ... --dynamic ... --motor DM-J4340 --model stribeck`): fits friction and inertia, writes `act_<motor>_params.json` and the curve/torque plots. Add `--gravity-split` if a load was present.
4. **Export** (`act_03_export_mujoco.py --params ... --joint r_elbow`): prints the MuJoCo joint attributes and a feed-forward friction-compensation table.
5. **Apply.** Paste `frictionloss`/`damping`/`armature` onto the matching joint in `mabel_full.xml`; add $\tau_f(\dot{q})$ as feed-forward to the controller's existing inverse-dynamics torque:

$$\tau_{ff} = \underbrace{\texttt{arm.gravity_torque(q)}}_{\texttt{mabel_sysid}} + \underbrace{\texttt{friction_torque(params, qd)}}_{\text{this package}}.$$

Safety. Keep clear of the shaft and keep a hand on the e-stop. The default speeds are a conservative fraction of rated v_{max} and the chirp amplitude is small; raise them only after a low-speed dry run. Verify each motor's $(p_{max}, v_{max}, t_{max})$ in the Damiao desktop tool match `src/mabel_actid/motors.py` — the MIT protocol packs floats using exactly these ranges, so a mismatch silently rescales every value on the wire.

7 Experimental results (bench)

This section reports the calibrations actually run on the MABEL bench. Two physical Damiao actuators were identified on a SocketCAN bus (`can0`, 1 Mbit/s, OpenMoko *gs_usb* adapter): a `DM-J4340P` and a `DM-J4310`. All raw logs, plots and parameter files live under `results/<motor>/`.

7.1 Protocol scaling must be verified first

The MIT protocol packs each quantity into a fixed-width integer using the motor's programmed range; the host decode is the inverse affine map

$$x = \frac{n}{2^b - 1}(x_{max} - x_{min}) + x_{min}, \qquad x_{min} = -x_{max}, \tag{9}$$

Table 1: Bench configuration.

item	value	note
bus	SocketCAN `can0` @ 1 Mbit/s	gs_usb adapter
control mode	MIT (impedance)	onboard sensored FOC
torque source	motor's CAN feedback	output-side, $i_q \cdot K_t$
excitation	position-ramp staircase + chirp	§6
loop rate	500 Hz	busy-wait scheduler

so a host range $x_{\max}^{\text{host}}$ that differs from the on-motor $x_{\max}^{\text{mot}}$ rescales *every* sample by $x_{\max}^{\text{host}}/x_{\max}^{\text{mot}}$. We detect this in situ by comparing the velocity read back over CAN to the position-derived velocity $\dot{q} = \mathrm{d}q/\mathrm{d}t$ (which depends only on $p_{\max}$): their ratio is exactly the velocity-range error. Two motors were initially mislabeled; the ratio test corrected both (Table 2).

Table 2: Velocity-range correction from the $\dot{q}$ ratio test. The "readback/$\dot{q}$" column is the measured scaling error with the assumed $v_{\max}$.

unit (CAN id)	assumed $v_{\max}$	readback/$\dot{q}$	true $v_{\max}$	also corrected
"J4310" (id 1)	30 rad/s	3.02	10.4 rad/s	is a `DM-J4340P`, $t_{\max}$ 10 → 27
`DM-J4310` (id 2)	30 rad/s	1.10	≈27 rad/s	—

The torque range $t_{\max}$ has no in-situ reference, so it must be read from the desktop tool: correcting the mislabeled unit from $t_{\max} = 10$ to 27 N m scales all of its friction torques by $27/10 = 2.7$.

7.2 Identified parameters

Table 3 collects the fits. The simulation self-test (§5) recovers the controller-relevant terms F_c, F_v, J to the noise floor from a known ground truth; the two bench motors are then identified the same way. The friction model fits the constant-velocity points with $R^2 > 0.99$ in every case.

Table 3: Identified friction and reflected-inertia parameters. "sim" lists ground truth → identified. $F_s = F_c$ where the break-away was not separable (low-speed stick-slip). J for the `DM-J4310` is pending (its chirp aborted, §7.4).

parameter	unit	sim (true → id.)	`DM-J4340P`	`DM-J4310`
F_c^+	N m	0.450 → 0.450	0.424	0.079
F_c^-	N m	0.520 → 0.519	0.436	0.087
F_v	N m s/rad	0.080 → 0.081	0.212	0.0198
F_s^+	N m	0.700 → 0.450	0.424	0.108
F_s^-	N m	0.780 → 0.726	0.436	0.129
v_s	rad/s	0.150 → 0.165	0.180	0.279
J	kg m^2	0.0120 → 0.0120	0.0180	—
friction R^2	—	0.931	0.9992	0.9933
friction RMS	N m	0.0024†	0.0187	0.0073†

†inlier RMS (excludes the unmodelled stick-slip points).

Table 4: MuJoCo joint attributes (paste onto the matching joint in `mabel_full.xml`). `frictionloss` is the direction-averaged F_c.

motor	`frictionloss`	`damping`	`armature`
`DM-J4340P`	0.429852	0.211518	0.0179936
`DM-J4310`	0.082864	0.019760	(pending)

7.3 Friction curves and fits

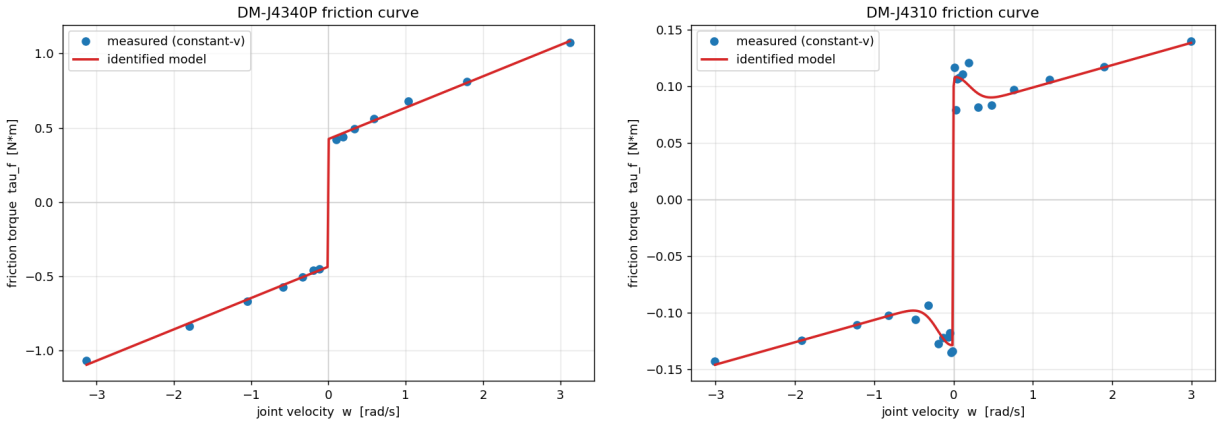


Figure 1: Measured constant-velocity points (markers) and identified Coulomb + viscous + Stribeck model (line). Left: `DM-J4340P` — a clean curve, sharp break-away at $F_c \approx 0.43\,\mathrm{N\,m}$. Right: `DM-J4310` — lighter friction ($F_c \approx 0.08\,\mathrm{N\,m}$) but visibly noisier (more torque ripple), with a genuine Stribeck dip ($F_s > F_c$).

7.4 Two methodology findings

Position-ramp staircase. A pure velocity hold ($k_p = 0$) cannot break the joint through stiction at low speed — it sticks at $\omega \approx 0$ and the Stribeck region is lost. Commanding instead a position ramp $q_{\mathrm{ref}}(t) = q_0 + \omega\, t$, *re-anchored to the measured position at the start of each level*, forces a clean constant velocity at every step without the overshoot a globally-integrated ramp accrues from carried-over position error.

Bus robustness. The aggressive chirp ($k_p = 40$) overcurrent-faulted the smaller `DM-J4310`; it stopped acknowledging and the 500 Hz stream overflowed the default 10-frame SocketCAN tx queue (`ENOBUFS`). Mitigations now in place: a bounded retry in the sender, `txqueuelen` raised to 1000, and gentler chirp defaults for small motors. The staircase (hence the full friction fit) completed before the fault; only J is outstanding.

8 Notes and extensions

- **Temperature & load dependence.** Friction drifts with temperature (warm-up) and gearbox load. Calibrate warm, and if your duty cycle varies widely consider a small load-dependent term.

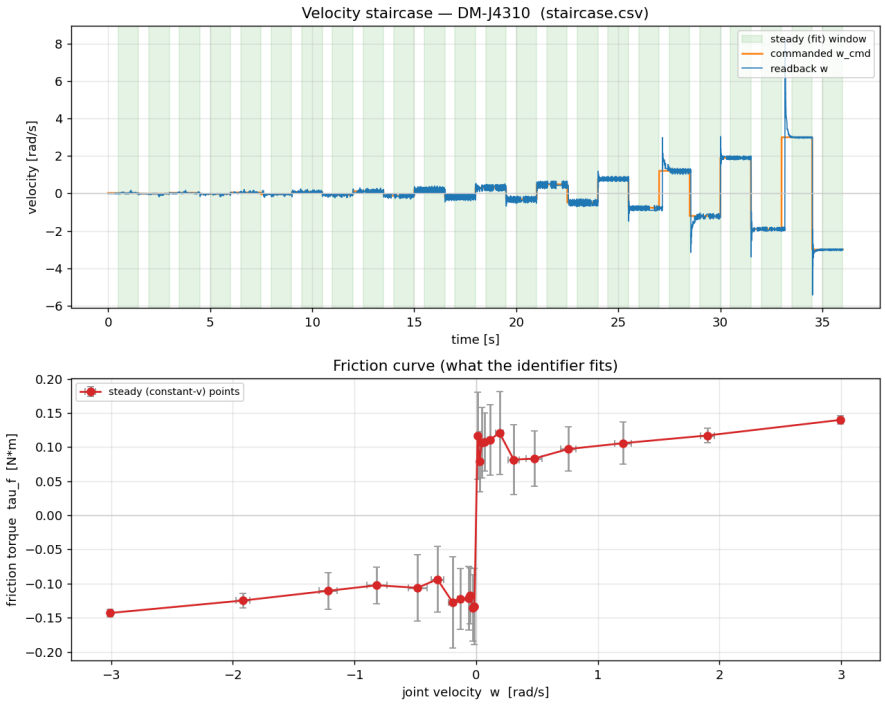


Figure 2: `DM-J4310` velocity staircase. Top: the readback velocity (blue) tracks the command (orange) at every level once $v_{\max}$ is corrected; shaded bands are the settled windows averaged for the friction curve. Bottom: the resulting $\tau_{\mathrm{f}}(\omega)$.

- **Position dependence.** Cogging and gear-mesh ripple give a small position-periodic torque; if it matters, add a Fourier-in-q term to the regression.
- **In-situ (on-robot) identification.** Equation (5) holds per joint on the assembled arm if $\tau_{\mathrm{g}}(q)$ is the identified gravity model from `mabel_sysid`; subtract it and the same staircase/chirp fit applies, letting you recalibrate without a bench.